

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2023

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
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- Let G be a group and let Z be the center of G (that is, Z = the elements that commute with everything in G).
 - Show that if G/Z is cyclic then G is abelian.
 - Let $\text{Aut}(G)$ be the group of automorphisms of G . Show that if $\text{Aut}(G)$ is cyclic then G is abelian.
- Let A and B be complex $n \times n$ matrices. Then $A \otimes I$ acts on $\mathbb{C}^n \otimes_{\mathbb{C}} \mathbb{C}^n$ via $(A \otimes I)(v \otimes w) = Av \otimes w$, and similarly for $I \otimes B$. Show that $A \otimes I - I \otimes B$ is invertible if and only if the eigenvalues of A are distinct from the eigenvalues of B .
- Let R be a commutative ring with identity and let J be the intersection of all maximal ideals of R . Let $r \in R$. Show that $r \in J$ if and only if $1 - ry$ is a unit in R for all $y \in R$.
- Let R be a commutative ring with 1 and let M be an R -module. Suppose we have exact sequences of R -modules

$$0 \rightarrow K_1 \rightarrow P_1 \xrightarrow{\phi_1} M \rightarrow 0$$

and

$$0 \rightarrow K_2 \rightarrow P_2 \xrightarrow{\phi_2} M \rightarrow 0$$

where P_1 and P_2 are projective.

- Let $X = \{(x, y) \in P_1 \times P_2 \mid \phi_1(x) = \phi_2(y)\}$, and let $\pi_1 : X \rightarrow P_1$ be the projection onto the first component. Show that π_1 is surjective and its kernel is isomorphic to K_2 .
- Show that $X \simeq K_2 \oplus P_1$.
- Show that $K_2 \oplus P_1 \simeq K_1 \oplus P_2$.

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5. Let $\alpha \in \mathbb{C}$ be a root of $f(x) = x^6 + 3$, and let $K = \mathbb{Q}(\alpha)$.
- (a) Show that K contains the 6th root of unity $\omega = (1 + \sqrt{-3})/2$.
 - (b) Show that K is the splitting field of $f(x)$.
 - (c) Determine $\text{Gal}(K/\mathbb{Q})$. (*Hint:* Show that K contains $3^{1/3}$.)
6. Let G be a finite group and let A be a subgroup of G with $[G : A] = m$. Let $G = g_1A \cup \cdots \cup g_mA$ be a coset decomposition of G with respect to A . Let $\rho : G \rightarrow \text{GL}(V)$ be an irreducible representation of G on a complex vector space V .
- (a) Let W be a nonzero subspace of V such that $\rho(a)W = W$ for each $a \in A$. Show that $\rho(g_1)W + \cdots + \rho(g_m)W = V$.
 - (b) Suppose A is abelian. Show that $\dim V \leq m$.